

# BILLEL ZIGHAB

## Senior Solution Architect

Enterprise & Solution Architecture · Cloud / On-Premise / Hybrid · Standards, Governance & Assurance

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English · French · Arabic — Fluent

### PROFESSIONAL SUMMARY

Senior Solution Architect, 17+ years of IT experience including 10+ years of end-to-end architecture on critical, high-volume digital platforms. Designs large-scale distributed and transactional architectures and defines applicative and cloud evolution trajectories (target + transition) in international contexts. Specialist in microservices architectures, event-driven decoupling (EDA, CQRS) and API Design / API Management strategies (gateway, contracts, versioning). Drives the urbanization of complex systems and supports teams on cross-system integration choices, challenging technical decisions and making the right performance / cost / time trade-offs. Owns application security by design: authentication, authorization and token management (OAuth2/OIDC, JWT, mTLS), IAM / CIAM, Zero Trust. Strong command of AWS and working knowledge of GCP, hybrid / multi-cloud architectures, high availability, resilience and data integrity. Also designs the integration architecture for AI/ML capabilities developed by data science teams. Recognized for explaining complex topics to business and technical stakeholders and uniting teams in Agile / multicultural settings.

### TECHNICAL SKILLS

#### Frameworks, Governance & Architecture Deliverables

TOGAF, ArchiMate, DAT, HLD, LLD, ADR, C4 models, flow & impact matrices, DR/BCP dossier, DAG, architecture diagrams, data model, reference patterns, API contracts (OpenAPI)...

#### Application & Data Architecture Patterns

Microservices, EDA, CQRS/Event Sourcing, Saga, API Gateway & BFF, Strangler Fig, Data Mesh, Lakehouse, Lambda/Kappa Architecture, DDD, Polyglot Persistence

#### Cloud (Azure / AWS)

AKS, APIM, Databricks, Synapse, Event Hubs, Functions, Key Vault, Front Door, WAF, ADLS Gen2, Cosmos DB · EKS, Lambda, MSK, SageMaker, API GW, S3, Redshift · Terraform, Bicep, Helm, Kubernetes, Docker

#### Security & Compliance

Zero Trust, Defense in Depth, IAM/Federation, mTLS, PKI, Key Management, OWASP, GDPR, IRM, SAML, JWT, PKCE, OAuth2/OIDC

#### Modeling

BiZZdesign Enterprise Studio, SAP PowerDesigner, ArchiMate, C4/Structurizr, Visio, Lucidchart

#### Dev & DevSecOps

Java/Spring Boot, Python, Scala, REST/OpenAPI, Angular, React · GitLab CI, Jenkins, SonarQube, Checkmarx, Artifactory

#### Data

Oracle, PostgreSQL, SQL Server, MongoDB, Cassandra, Elasticsearch, Cloudera/HDP (Spark, HDFS, Hive)

#### AI/ML — Architecture & Integration

Integration architecture for ML platforms within the enterprise IS: Computer Vision (CNN, EfficientNet), time-series forecasting (LSTM, XGBoost, Prophet), MLOps (MLflow, SageMaker, Model Registry), TensorFlow/Keras — interface contracts, data & security governance, feasibility ADRs.

### PROFESSIONAL EXPERIENCE

#### Shell Trading & Supply — NBS · Carbon Management · Power Portfolio

Feb 2022 – Present

Senior Solution Architect · Greater London, United Kingdom · Hybrid

- Led the end-to-end architecture of 3 mission-critical portfolios (Nature-Based Solutions, Carbon Management, Flex GB/Batteries) spanning 30 multi-region Azure applications, from business-requirements capture to production, with a 99.9% + SLA and a disaster-recovery plan sized for RTO < 1h / RPO < 15 min.
- Designed a real-time event-driven architecture (Spark Structured Streaming, CQRS, Saga Pattern) for the Flex GB battery trading platform, reducing Day-Ahead decision latency by 40%.
- Designed the integration architecture for the long-term carbon emissions predictive modeling platform (2050 horizon, NBS/Carbon Management portfolios) developed by the data science team: defined interface contracts, MLOps requirements (versioning/monitoring via Azure Databricks/MLflow), security and governance of satellite data (NDVI, biomass), and guaranteed the feasibility and integrity of the multi-model ML solution (LSTM, XGBoost, spatial regression) within the enterprise IS for Net Zero 2050 carbon-offsetting decision support.

- Formalized the full architecture documentation chain — conceptual, logical and technical views, HLD, LLD and ADRs — and defended the dossiers before the Architecture Review Board (ARB) and Design Authority, cutting architecture-approval time by 35% for 8 product teams while ensuring IRM, GDPR and Security by Design compliance.
- Reduced cloud infrastructure costs by 25% through shared landing zones and dynamic AKS/Databricks scaling policies, while enforcing data encryption at rest and in transit (TLS/mTLS, Key Vault) across all flows.
- Mentored 20 senior architects and engineers, contributed to the group Technology Radar and led platform cost-optimization initiatives in collaboration with the SOM and Enterprise Architecture teams.

*Deliverables* : DAT (Technical Architecture Document), HLD, LLD, ADR, C4 models, flow & impact matrices, blueprints, DR/BCP plan.

*Tech stack* : Azure (AKS, ADF, Databricks, MLflow, Event Hubs, Spark Structured Streaming, Synapse, APIM, Service Bus, Functions, Key Vault, Front Door, App Gateway, WAF, ADLS Gen2, Azure SQL, Cosmos DB, Log Analytics, Azure DevOps), TensorFlow/Keras, XGBoost, Python/Scikit-learn, Terraform, Bicep, Endur, Aligner, BiZZdesign, SAP PowerDesigner.

## EDF / ENEDIS — Data Department (via B4ALL Consulting)

Jul 2020 – Feb 2022

**Senior Solution Architect** · Paris, France · Agile / SAFe

- Designed the cloud-native, distributed architecture of the Data IS (AWS, Cloudera/HDP, Kafka), from business-requirements capture to target architectures, aligned with group standards (Cloud First, Open-Source First).
- Defined applicative and cloud evolution trajectories (target + transition) and drove the modernization of 7 legacy IS components, reducing technical debt by 30% and improving the scalability of high-volume pipelines.
- Designed inter-application and integration architectures — interface contracts, exchange formats, protocols, resource sizing — and qualified data sensitivity (CIA, GDPR) with business and security teams.
- Created 12 reusable architecture patterns and reference models adopted by 5 teams (time-to-design -30%), and challenged technical choices in the architecture committee (licensing costs -40%).
- In addition, to cover a temporary shortage of data science resources, designed the integration architecture and produced the dossier for a Computer Vision solution (CNN EfficientNet-B4, model built with the data team) for Linky meter quality control (28M meters), deployed on AWS (SageMaker, API Gateway, Lambda, S3) and consumed by the ENEDIS technicians' mobile app (94% F1-score validated).

*Deliverables* : DAT, HLD/LLD, ADR, interface contracts, data model, evolution trajectory (target + transition), reference patterns.

*Tech stack* : Cloudera/HDP (Spark, Sqoop, HDFS, Hive, HBase, Hue), Kafka, AWS (SageMaker, Lambda, API Gateway, S3, ECR, CloudWatch), TensorFlow/Keras (EfficientNet, transfer learning), Python (OpenCV, Pillow, NumPy, Pandas, Scala), Oracle, Teradata, PostgreSQL, MariaDB, Neo4J, Tableau, Power BI, Dataiku, REST, Microservices, Spring Boot, Kubernetes, Docker, GitLab, Jenkins, Artifactory/Nexus, SonarQube, Checkmarx, Tock Chatbot.

## Europcar Mobility Group — Click & Go

Jan 2019 – Jun 2020

**Solution Architect** · Paris, France · Agile / Scrum

- Conducted a full architectural audit of the Click & Go platform (B2B & B2C portal + Android/iOS mobile app), covering backend, frontend, mobile and integration layers — identified 15 critical technical-debt points and defined a refactoring roadmap prioritized over 4 quarters, validated by group technical management.
- Designed the centralized API Gateway architecture (Kong) and the backend microservices decomposition, with explicit interface contracts and TLS encryption of B2B/B2C flows, reducing inter-service coupling by 60% and time-to-release by 45% (18 independently deployable components).
- Defined the platform's cross-cutting security architecture (Spring Security, Spring LDAP, OAuth2) for B2B (partner access) and B2C (general public) authentication flows, ensuring compliance with group IS policies and GDPR requirements.
- Designed solution components and architecture models for the Spring Batch (bulk reservation processing) and Spring Data JPA (Oracle/PostgreSQL access) layers, producing HLD/LLD deliverables, technical analyses and ADRs for governance reviews.
- Implemented the complete CI/CD pipeline (Jenkins, SonarQube, Artifactory, GitLab, Newman) automating API integration tests and reducing regressions by 45% — supported tech leads and development teams during implementation phases and architecture decisions.

*Deliverables* : DAT, HLD, LLD, ADR, API contracts (OpenAPI), security dossier, technical analyses.

*Tech stack* : Java 11, Spring Boot 2.x, Spring Security, Spring Batch, Spring MVC REST, Spring LDAP, Spring Data JPA, Microservices, Kong API Gateway, Oracle, PostgreSQL, Business Objects, Android & iOS, Docker, OpenShift, Angular 7, Bootstrap, Jenkins, SonarQube, Artifactory, GitLab, Newman, IntelliJ.

## RTE — Réseau de Transport d'Électricité

May 2018 – Jan 2019

**Technical Solution Architect** · Paris, France · Agile / Scrum

- Established the reference technical architecture of the GUIDE platform (50,000+ external identities) — Spring Boot + OpenLDAP + Angular — and authored the architecture vision documents (application, global, functional), delivered on time with 0 critical defects in production.
- Trained and coached 12 developers in DDD, TDD, CI and design-pattern practices, reducing the defect rate by 40% within the first 3 sprints post-coaching.

*Deliverables* : Architecture vision documents (application, global, functional), DAT, HLD/LLD.

*Tech stack* : Java 8, Spring Boot 2.x, Spring Security, Spring LDAP, REST APIs, Angular 6, Docker, Jenkins, SonarQube, OpenLDAP, Maven 3.

## AFPA

Jan 2017 – Apr 2018

**Application & Technical Architect / Technical Lead** · Paris, France · Scrum

- Defined the conceptual and technical architecture of CIRUS (data model, microservices/API layer, Angular, security, observability) and produced the HLD/LLD, presented to and validated by the enterprise architecture committee.

- Accelerated the delivery of 5 critical modules by directly developing 5 components alongside the team, cutting delivery time by 3 sprints on a 6-quarter plan.

*Tech stack : Java 8, Spring Boot, Spring Security, Hibernate, Docker, Angular 4, SQL Server 2012, MySQL, Jenkins, SonarQube, GitLab.*

## ENEDIS — OpenData

Apr 2016 – Dec 2016

**Big Data Technical Lead** · Paris, France · Scrum

- Designed and delivered the OpenData large-scale data publication and processing pipeline (Spark, Hadoop), processing 30 public datasets with 99% availability.
- Coached 6 junior developers in Big Data practices and Git branching, reducing the integration error rate by 60% over the last 2 sprints.

*Tech stack : Java 8, Spark, Hadoop, Hive, HDFS, Docker, Teradata, Jenkins, SonarQube.*

## Airbus Helicopters — ORION

Jul 2013 – Mar 2016

**Technical Lead & Junior Technical Architect** · Marseille, France · Agile / Scrum

- Harmonized and industrialized 18 application components on the ORION platform, reducing code-duplication debt by 35% and improving the maintainability of documentation and maintenance manuals.
- Developed an API health-check system reducing MTTR (Mean Time To Recovery) by 60% by detecting service failures before user impact. Deployed full DevOps automation (Jenkins, Nexus, Sonar, Git).

*Tech stack : Java/JEE, REST, AngularJS, Angular 2, XML/XSLT/XPath, Spring Boot, Docker, Jenkins, Nexus, SonarQube, SQL Server, MySQL, Active Directory LDAP, IntelliJ, Maven 3.*

## SNCF — Portail Affilié CPRP

Oct 2011 – Jun 2013

**Senior Java/JEE Developer** · Marseille, France · Agile

- Designed and developed the complete architecture of the CPRP affiliate portal (Java/JEE, WebLogic, Oracle 11g, REST, Liferay 6.2) in 8 months, delivered on time with 7,000 affiliated users migrated without service interruption.

*Tech stack : Java/JEE, EJB 3, Liferay 6.2, Hibernate, JSF, REST (RESTEasy), Oracle 11g, Maven, Tomcat 7, Git.*

## Direction Générale des Finances Publiques

Feb 2011 – Sep 2011

**Talend & Senior Java Developer** · Paris, France · Agile

- Developed and delivered the note-sharing and financial-control system in 7 months. Successfully executed the MySQL → PostgreSQL and Drupal Commons 6 → 7 data migration with 0 data loss across 600,000 records.

*Tech stack : Java 6, Talend Studio, Hibernate, Spring, Maven 2, PrimeFaces, MySQL, PostgreSQL.*

## Arts et Métiers ParisTech

Jul 2009 – Jan 2011

**Java/JEE Developer** · France · Agile

- Delivered the critical business interfaces between the KAKI payroll module and the Concerto Adix accounting system, ensuring zero-defect integration of payroll for 1,200 employees into the ENSAM financial system.

*Tech stack : Java/JEE, Spring MVC, Spring Security, Oracle 10g, JasperSoft, Apache POI, PL/SQL, JBoss, Maven, Ant, Selenium.*

## Ville de Toulon & Métropole TPM

Jul 2008 – Jun 2009

**PHP Developer** · Toulon, France · Agile

- Rebuilt and delivered the toulon.fr & tpm-agglo.fr websites on Drupal 7 — 8 custom modules developed, 120K visitors/month supported after go-live.

*Tech stack : Drupal 7, PHP, HTML5, CSS, jQuery, Apache 2, MySQL, Solr, Git.*

## Université Aix-Marseille — Saint-Jérôme Faculty Library

Jan 2008 – Jun 2008

**Java/JEE Developer** · Aix-Marseille, France · Agile

- Rebuilt the library's document management system (Java/JEE, Spring, Hibernate, SOAP), delivered and deployed across 3 environments (test, training, production) within 6 months.

*Tech stack : Java/JEE, Spring, Hibernate, JSF, PrimeFaces, Tomcat 6, SOAP/WSDL, MySQL, JasperReports, SVN.*

## EDUCATION

### Master's in Distributed Systems

2008

Université Aix-Marseille, Aix-Marseille, France

## LEADERSHIP & DISTINCTIONS

- Mentored 8+ senior architects and engineers at Shell, contributed to the group Technology Radar and led cloud cost-optimization initiatives — a cross-cutting role recognized beyond the direct project scope.
- Evangelized modern engineering practices (DDD, TDD, CI, Zero Trust, Security by Design) across 12+ development teams at RTE, AFPA and Airbus Helicopters over 3 consecutive assignments.
- Trilingual English / French / Arabic (fluent) — facilitator of collaboration in international and cross-cultural multi-team contexts (Shell UK, EDF, Europcar, DGFIP).
- Presented and defended 35 architecture proposals before enterprise architecture committees and governance bodies (Shell EA Board, AFPA Architecture Committee, RTE Architecture Review).